Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Do Integrins Mediate the Skeletal Responses to Altered Loading In Vivo?		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	10/01/2000	**End Date:**	09/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Our long-term objective is to define the molecular events that mediate skeletal adaptations to changes in weightbearing, as occurs during and after space flight. The working hypothesis of this proposal is that intact integrin/extracellular matrix interactions in osteoblasts are required for bones to adapt appropriately to changes in mechanical loading. We will determine if transgenic mice, which express a dominant-negative beta1 integrin targeted to osteoblasts, can adapt to changes skeletal loading similarly to wild-type mice. Skeletal loading will be manipulated in vivo using the rodent hindlimb unloading model, since it selectively prevents weightbearing of the hindlimbs and normal loading can subsequently resume. In Aim 1, we will test hypothesis that disruption of integrin function in osteoblasts in vivo impairs skeletal adaptations to disuse. In Aim 2, we will test the hypothesis that disruption of beta1 integrin function in osteoblasts impairs their ability to adapt to increased mechanical loading. This project uses a molecular approach to gain insight into the cellular basis of mechanotransduction, and will provide the first in vivo test of the importance of integrins during skeletal loading. This project will also benefit NASA by establishing a novel model to evaluate		

	mechanisms leading to skeletal recovery from the adverse effects of disuse.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is basic science, designed to unravel the molecular mechanisms underlying normal and abnormal processes in bone. Successful completion of these experiments will define critical molecular components of the pathways regulating bone mass in response to mechanical stimuli, such as weightbearing. This will provide novel insight into mechanisms of disuse osteoporosis caused by immobilization and bed rest, as well as spaceflight. Humans, as well as mice, have the adhesion proteins, integrins, on their cell surface. Mice are unlikely to have unique and completely distinct mechanisms from humans for responding to mechanical forces that are generated by weightbearing. Identification of these molecular components will potentially help identify potentially useful countermeasures. Thus, although our study does not directly answer the question of how the human skeleton responds to the challenge of spaceflight, we will learn more about fundamental mechanisms with the ultimate potential of benefiting the health of astronauts and others on Earth.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 06/04/2025)